

Program 7 – Options Menu

res/menu/menus.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/php"
        android:title="PHP"/>
    <item
        android:id="@+id/java"
        android:title="JAVA"/>
    <item
        android:id="@+id/csharp"
        android:title="C#"/>
</menu>
```

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu Example"
        android:textSize="24sp"
        android:textStyle="bold" />

</LinearLayout>
```

java/.../MainActivity.java

```
package com.example.lab7;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreatePanelMenu(int featureId, @NonNull Menu menu) {
        MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.menus, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        if (item.getItemId() == R.id.php) {
            Toast.makeText(this, "Php Page", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.java) {
            Toast.makeText(this, "Java Page", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.csharp) {
            Toast.makeText(this, "C# Page", Toast.LENGTH_SHORT).show();
        }
        return super.onOptionsItemSelected(item);
    }
}
```

```
}  
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<manifest xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:tools="http://schemas.android.com/tools">  
  
    <uses-permission android:name="android.permission.INTERNET" />  
  
    <application  
        android:allowBackup="true"  
        android:dataExtractionRules="@xml/data_extraction_rules"  
        android:fullBackupContent="@xml/backup_rules"  
        android:icon="@mipmap/ic_launcher"  
        android:label="@string/app_name"  
        android:roundIcon="@mipmap/ic_launcher_round"  
        android:supportRtl="true"  
        android:theme="@style/Theme.AppCompat.Light"  
        tools:targetApi="31">  
        <activity  
            android:name=".MainActivity"  
            android:exported="true">  
            <intent-filter>  
                <action android:name="android.intent.action.MAIN" />  
                <category android:name="android.intent.category.LAUNCHER" />  
            </intent-filter>  
        </activity>  
    </application>  
  
</manifest>
```

Program 8 – File Storage (Read / Write)

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/editTextInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:autofillHints="text"
        android:hint="@string/enter_text_here"
        android:inputType="text"
        android:padding="10dp"
        android:textSize="16sp" />

    <Button
        android:id="@+id/btnSave"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/save"
        android:layout_marginTop="15dp" />

    <Button
        android:id="@+id/btnRead"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/read"
        android:layout_marginTop="10dp" />

    <TextView
        android:id="@+id/textViewOutput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/output_will_appear_here"
        android:textSize="18sp"
        android:layout_marginTop="20dp"
        android:padding="10dp" />

</LinearLayout>
```

java/.../MainActivity.java

```
package com.example.labprogram8;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class MainActivity extends AppCompatActivity {

    private static final String FILE_NAME = "example.txt";
    EditText mEditText;
    TextView mTextView;
    Button btnSave, btnRead;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mEditText = findViewById(R.id.editTextInput);
        mTextView = findViewById(R.id.textViewOutput);
        btnSave = findViewById(R.id.btnSave);
```

```

        btnRead = findViewById(R.id.btnRead);

        btnSave.setOnClickListener(v -> saveFile());
        btnRead.setOnClickListener(v -> readFile());
    }

    public void saveFile() {
        String text = mEditText.getText().toString();
        try (FileOutputStream fos = openFileOutput(FILE_NAME, MODE_PRIVATE)) {
            fos.write(text.getBytes());
            mEditText.getText().clear();
            Toast.makeText(this, "Saved successfully!", Toast.LENGTH_SHORT).show();
        } catch (IOException e) {
            e.printStackTrace();
            Toast.makeText(this, "Error saving file", Toast.LENGTH_SHORT).show();
        }
    }

    public void readFile() {
        try (FileInputStream fis = openFileInput(FILE_NAME)) {
            int c;
            StringBuilder sb = new StringBuilder();
            while ((c = fis.read()) != -1) {
                sb.append((char) c);
            }
            mTextView.setText(sb.toString());
            Toast.makeText(this, "Data loaded successfully", Toast.LENGTH_SHORT).show();
        } catch (IOException e) {
            mTextView.setText("File not found");
            Toast.makeText(this, "File not found", Toast.LENGTH_SHORT).show();
            e.printStackTrace();
        }
    }
}

```

Program 10 – Send SMS

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editText1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Phone Number"
        android:inputType="phone" />

    <EditText
        android:id="@+id/editText2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Message"
        android:inputType="textMultiLine" />

    <Button
        android:id="@+id/btnSendSms"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send SMS" />

</LinearLayout>
```

java/.../MainActivity.java

```
package com.example.lab10;

import android.os.Bundle;
import android.Manifest;
import android.annotation.SuppressLint;
import android.content.pm.PackageManager;
import android.telephony.SmsManager;
import android.widget.*;
import android.view.*;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

public class MainActivity extends AppCompatActivity {

    Button btn1;
    EditText et1;
    EditText et2;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ActivityCompat.requestPermissions(this,
            new String[]{Manifest.permission.SEND_SMS},
            PackageManager.PERMISSION_GRANTED);

        btn1 = findViewById(R.id.btnSendSms);
        et1 = findViewById(R.id.editText1);
        et2 = findViewById(R.id.editText2);

        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String number = et1.getText().toString();
                String sms = et2.getText().toString();
                try {
                    SmsManager smsManager = SmsManager.getDefault();
                    smsManager.sendTextMessage(number, null, sms, null, null);
                    Toast.makeText(getApplicationContext(),
```

```

        "Message sent successfully", Toast.LENGTH_SHORT).show();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
});
}
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="sms"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Lab10">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

```

Program 11 – Send Email (Intent)

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/et1_emailId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Receiver Email"
        android:inputType="textEmailAddress"/>

    <EditText
        android:id="@+id/et2_emailSub"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Subject"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/et3_emailMsg"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"
        android:layout_marginTop="15dp"
        android:lines="5"
        android:gravity="top"/>

    <Button
        android:id="@+id/btn_send"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send Email"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

java/.../MainActivity.java

```
package com.example.lab11;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button button;
    EditText sendto, subject, body;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        sendto = findViewById(R.id.et1_emailId);
        subject = findViewById(R.id.et2_emailSub);
        body = findViewById(R.id.et3_emailMsg);
        button = findViewById(R.id.btn_send);

        button.setOnClickListener(view -> {
            String emailsend = sendto.getText().toString().trim();
            String emailsubject = subject.getText().toString().trim();
            String emailbody = body.getText().toString().trim();

            if (emailsend.isEmpty()) {
                Toast.makeText(this, "Please enter recipient email",
```

```
        Toast.LENGTH_SHORT).show();
        return;
    }

    Intent intent = new Intent(Intent.ACTION_SEND);
    intent.putExtra(Intent.EXTRA_EMAIL, new String[]{emailsend});
    intent.putExtra(Intent.EXTRA_SUBJECT, emailsubject);
    intent.putExtra(Intent.EXTRA_TEXT, emailbody);
    intent.setType("message/rfc822");

    if (intent.resolveActivity(getPackageManager()) != null) {
        startActivity(Intent.createChooser(intent, "Choose an Email client:"));
    } else {
        Toast.makeText(this, "No email app found", Toast.LENGTH_SHORT).show();
    }
}

});
}
```